

ASSIGNMENT 1

1. Difference between Frontend, Backend and Full-Stack Development

Frontend Development

Frontend is the part of a website that users can see and use. It includes buttons, images, text, menus, etc. It is made using HTML, CSS and JavaScript.

Example: The homepage of Amazon or Flipkart.

Backend Development

Backend is the part that works behind the scenes. It stores data, checks login details, and connects with the database.

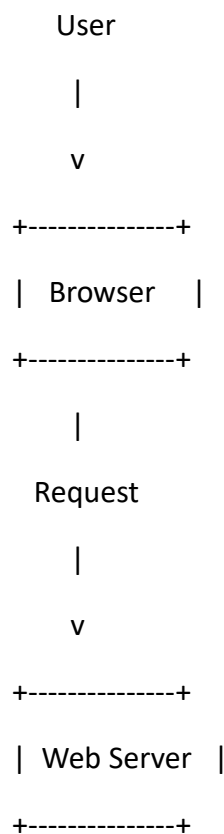
Example: When we log in to Instagram, the backend checks our username and password.

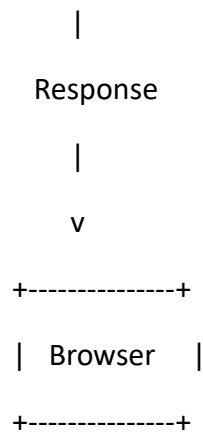
Full-Stack Development

A full-stack developer works on both frontend and backend. They can build a complete website from start to finish.

Example: A developer creating an online shopping website with login, products and payment system.

2. Client-Server Model





3. How a Browser Displays a Web Page

1. The user enters a website address in the browser.
2. The browser sends a request to the web server.
3. The server receives the request.
4. The server finds the required web page.
5. It sends HTML, CSS and JavaScript files back.
6. The browser reads these files.
7. Finally, the web page is displayed on the screen.

4. Tools Required for Web Development

Tool	Purpose
VS Code	Writing HTML, CSS and JavaScript code
Google Chrome	Testing the website
Live Server	Opens the website and refreshes automatically
Git	Keeps track of code changes
GitHub	Stores projects online
Node.js	Runs JavaScript outside the browser

5. What is a Web Server?

A web server is software that stores website files and sends them to users when they open a website.

Examples:

- Apache
 - Nginx
 - Microsoft IIS
 - LiteSpeed
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6. Roles in a Web Development Project

Frontend Developer

- Designs the website.
- Creates web pages.
- Makes the website user-friendly.

Backend Developer

- Writes server-side code.
- Connects the website with the database.
- Handles login and security.

Database Administrator (DBA)

- Creates and manages databases.
 - Keeps data safe.
 - Takes backups of data.
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7. Install VS Code

Steps:

1. Download VS Code.
2. Install it.
3. Install the Live Server extension.
4. Create a folder.

5. Create these files:
 - index.html
 - style.css
 - script.js
 6. Open **index.html** using Live Server.
 7. Take a screenshot and attach it to the assignment.
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8. Difference between Static and Dynamic Websites

Static Website

Content remains the same

Easy to build

No database

Faster

Dynamic Website

Content changes according to the user

More complex

Uses database

Slightly slower

Static Website Example: Personal Portfolio

Dynamic Website Example: Amazon, Facebook

9. Five Web Browsers and Their Rendering Engines

Browser

Rendering Engine

Google Chrome Blink

Microsoft Edge Blink

Mozilla Firefox Gecko

Safari WebKit

Opera Blink

Difference:

Different browsers use different rendering engines to display web pages. Because of this, some websites may look or work slightly differently in different browsers.

10. Basic Web Architecture

